MECHTRON / SFWRENG 3K04

Project Deliverable 1: Requirements Document

Part B: Hospital Monitoring System

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Group – 11

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Introduction

Our part of the project is creating a centralized human facing application which monitors and interacts with one or more ADCs. It is a program which interacts with the ADCs and provides a clinical and administrative overview. It will be used by nurses, doctors, pharmacists, administrators and technicians. The program will interact with the ADCs and will manage history, alarms, inventory, and cabinet health. The other part of the project that we are not in charge of has to do with the hardware for the individual cabinets, and direct cabinet control.

Functional Requirements (FR)

Principle	Code	Requirement
ADC Communication	FR-COM-##	01- The monitoring system shall establish and maintain communication with all configured ADCs in the hospital. 02- The monitoring system shall receive operational status, inventory, alarm, and event updates from connected ADCs. 03- The monitoring system shall associate each received communication with its source ADC, identified by cabinet number, floor, and wing.
Inventory Supervision	FR-INS-##	01- The monitoring system shall monitor medication quantities for each ADC and identify medications that are low in stock, out of stock, or require reorder. 02- The monitoring system shall notify within 1 hour to pharmacists and Administrative staff when medication requires refill or reorder and include the medication name, ADC number, location, and current quantity. 03- The monitoring system shall allow authorized users to view and update medication inventory information.
Alarm management	FR-ALM-##	01- The monitoring system shall receive, record, and display alarms from connected ADCs, including the source ADC, alarm type, severity, and timestamp. 02- The monitoring system shall identify security and operational alarms, including repeated failed authentication attempts, unauthorized

		access or break-in attempts, and drawers or doors left open beyond the configured time limit. 03- The monitoring system shall identify inventory and cabinet-health alarms, including low or out-of-stock medications, hardware faults, and communication failures. 04- The monitoring system shall route alarm notifications to the appropriate personnel within 1 hour based on alarm type, including pharmacists for medication-related alarms, technicians for hardware or communication faults, and administrators for security or operational alarms.
Historical Data	FR-HSD-##	01- The monitoring system shall store login logs for each user along with event data containing timestamp, ADC, medication and quantity dispensed. 02- The monitoring system shall store all ADC updates with timestamp and type of event for record 03- The monitoring system shall allow authorized users to retrieve historical dispensing event data. 04- The monitoring system shall allow users to filter data based on date, time, event and ADC
Reporting	FR-REP-##	01- The monitoring system shall allow authorized users to generate reports from stored historical data based on selected filters 02- The monitoring system shall generate inventory reports that identify medication stock levels and highlight medications requiring refill or reorder. 03- The monitoring system shall store report generation time stamp and user who requested the report and a copy of the report,
Operator dashboard	FR-OPD-##	01- The monitoring system dashboard shall display the operational status of each configured ADC on the operator dashboard. 02- The monitoring system dashboard shall display active alarms, urgency type, notified

		individual along with ADC and its location on the operator dashboard. 03- The monitoring system dashboard shall allow users to select an ADC or alarm to view more details about the flag and ADC
Cabinet Health Monitoring	FR-CHM-##	01- The monitoring system shall monitor the communication and hardware status of each configured ADC. 02- The monitoring system shall detect when an ADC stops sending expected status messages or reports a hardware fault and mark the affected cabinet unavailable. 03- The monitoring system shall restore the displayed cabinet status when communication or normal operation is re-established.

Non - Functional Requirements (NFR)

Principle	Code	Requirement
Safety	NFR-SAF-##	01- The monitoring system shall clearly distinguish critical alarms, warnings, unavailable ADCs, and normal ADC states to prevent safety-relevant conditions from being overlooked. 02- When communication with an ADC is lost, the monitoring system shall identify the cabinet as unavailable and clearly indicate that its displayed information may no longer be current. 03- An active alarm shall remain visible until it is acknowledged by an authorized user or the system receives confirmation that the alarm condition has been resolved.
Security	NFR-SEC-##	01- The monitoring system shall require users to authenticate before accessing protected system functions and information. 02- The monitoring system shall enforce role-based access control so that nurses, pharmacists, technicians, and administrators can access only the functions authorized for their assigned roles.

		03- The monitoring system shall record failed authentication attempts and generate a security alarm after a configured number of consecutive failed attempts. 04- The monitoring system shall automatically lock an authenticated session after a configured period of inactivity and require the user to authenticate again.
Reliability	NFR-REL-##	01- A communication or hardware failure involving one ADC shall not prevent the monitoring system from receiving, processing, or displaying information from other available ADCs. 02- The monitoring system shall preserve stored inventory, alarm, historical, and audit records following an application restart or unexpected shutdown. 03- After recovery from an interruption, the monitoring system shall restore its most recently stored state and resume monitoring configured ADCs.
Availability	NFR-AVA-##	01- The monitoring system shall remain operational for available ADCs when one or more other ADCs become unavailable. 02- The monitoring system shall automatically attempt to re-establish communication with an ADC after a temporary communication failure. 03- The monitoring application shall be accessible from authorized hospital workstations connected to the hospital network.
Performance	NFR-PER-##	01- Critical alarm information shall be displayed on the operator dashboard within 5 seconds of the monitoring system receiving the alarm from an ADC. 02- ADC operational status and inventory information shall be updated on the dashboard within 5 seconds of the monitoring system receiving a valid update. 03- Medication refill or reorder notifications shall be sent to the appropriate personnel within 1 hour of receiving an inventory update

		indicating that the configured threshold has been reached. 04- Requested ADC details and filtered historical records shall be displayed within 5 seconds under normal operating conditions.
Traceability	NFR-TRA-##	01- The monitoring system shall maintain an audit log of user logins, failed login attempts, inventory modifications, alarm acknowledgements, cabinet access events, and report generation. 02- Each audit record shall contain the event type, timestamp, associated user when applicable, and affected ADC identified by cabinet number and location when applicable. 03- Audit records shall not be modifiable or deletable through normal user credentials, only authorized user shall have access to do so.
Usability	NFR-USE-##	01- The dashboard shall use consistent visual indicators and text labels to distinguish normal, warning, critical, and unavailable ADC states. 02- Critical status information shall not be communicated using colour alone and shall also include a text label or visual symbol. 03- Alarm notifications shall display the alarm type, severity, source ADC, cabinet location, and timestamp without requiring the operator to open an additional view. 04- Error messages shall identify the failed operation and provide a description of the error when the cause is known.

User Stories

1. ADC Communication

US-COM-01 — Monitor Connected ADCs

User Story:
As a hospital operator, I want to receive updates from all connected ADCs with their cabinet number and location so that I can monitor cabinets throughout the hospital from one centralized system.

Traces to: FR-COM-01, FR-COM-02, FR-COM-03, NFR-PER-02

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
ADC Update Received (Normal)	An ADC is configured and connected to the monitoring system.	The ADC sends a status, inventory, alarm, or event update.	The monitoring system displays the received update and associates it with the correct ADC number, floor, and wing within 5 seconds.
ADC Communication Lost (Off-Normal)	An ADC is configured and was previously communicating with the monitoring system.	The monitoring system stops receiving the expected status messages from the ADC.	The ADC is displayed as unavailable and its displayed information indicates that it may no longer be current.
Unknown ADC (Off-Normal)	An ADC is not configured in the monitoring system.	The monitoring system receives communication from the unknown ADC.	The communication is not associated with an existing configured ADC, and the unknown communication event is recorded.

2. Inventory Supervision

US-INS-01 — Monitor Medication Stock

User Story:
As a pharmacist, I want to be notified when medication is low, out of stock, or requires reorder so that I can arrange replenishment before medication becomes unavailable.

Traces to: FR-INS-01, FR-INS-02, NFR-PER-03

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Refill Required (Normal)	An ADC contains a monitored medication with a configured refill or reorder threshold.	An inventory update indicates that the medication has reached the configured threshold.	The medication is displayed as requiring replenishment, and appropriate personnel receive a notification containing the medication name, current quantity, ADC number, and location within 1 hour.
Medication Out of Stock (Off-Normal)	An ADC contains a monitored medication.	The reported medication quantity reaches zero.	The medication is displayed as out of stock, and the appropriate personnel are notified.
ADC Communication Lost (Off-Normal)	Medication inventory information has previously been received from an ADC.	Communication with the ADC is lost.	The ADC is displayed as unavailable and its inventory information is identified as potentially outdated.

US-INS-02 — Manage Inventory Information

User Story:
As a pharmacist, I want to view and update ADC medication inventory information so that accurate medication stock information can be maintained.

Traces to: FR-INS-03, NFR-SEC-01, NFR-SEC-02, NFR-TRA-01

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Authorized Inventory Update (Normal)	An authenticated pharmacist has permission to modify inventory information.	The pharmacist submits a valid inventory update.	The updated inventory information is displayed and the modification is recorded in the audit log with the associated user.

Unauthorized Update (Invalid Access)	An authenticated user does not have permission to modify inventory information.	The user attempts to modify the inventory.	The modification is rejected and the inventory information remains unchanged.
Unauthenticated Access (Off-Normal)	A user has not authenticated with the monitoring system.	The user attempts to access protected inventory modification functions.	Access is denied and the user is presented with the authentication requirement.

3. Alarm Management

US-ALM-01 — View and Investigate Alarms

User Story:
As a hospital operator, I want to see active ADC alarms and their severity so that I can quickly identify conditions requiring attention.

Traces to: FR-ALM-01, FR-ALM-02, FR-ALM-03, NFR-PER-01, NFR-SAF-01, NFR-SAF-03

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Alarm Received (Normal)	A configured ADC is connected to the monitoring system.	The monitoring system receives an alarm from the ADC.	The alarm is displayed within 5 seconds with its type, severity, timestamp, source ADC, and location.
Security Alarm (Off-Normal)	An ADC reports repeated failed authentication attempts, unauthorized access, or a break-in attempt.	The monitoring system receives the security event.	The event is displayed as a security alarm with its source ADC, location, severity, and timestamp.
Alarm Not Yet Resolved (Off-Normal)	An active alarm is displayed.	The alarm has not been acknowledged and the system has not received confirmation that the condition is resolved.	The alarm remains visible as active on the operator dashboard.

Alarm Acknowledged (Normal)	An active alarm is displayed to an authorized user.	The authorized user acknowledges the alarm.	The alarm is displayed as acknowledged and the acknowledgement is recorded in the audit log with the associated user and timestamp.

US-ALM-02 — Route Alarm Notifications

User Story:
As a hospital operator, I want alarms to be routed to the personnel responsible for handling them so that the appropriate staff can respond to the problem.

Traces to: FR-ALM-04

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Medication Alarm (Normal)	A medication-related alarm has been generated.	The monitoring system routes the alarm notification.	The notification is sent to the appropriate pharmacist.
Hardware or Communication Alarm (Normal)	A hardware or communication fault alarm has been generated.	The monitoring system routes the alarm notification.	The notification is sent to the appropriate technician.
Security or Operational Alarm (Normal)	A security or operational alarm has been generated.	The monitoring system routes the alarm notification.	The notification is sent to the appropriate administrator.

4. Historical Data

US-HSD-01 — Review Historical Activity

User Story:
As an authorized hospital administrator, I want to review and filter historical ADC and user activity so that I can investigate previous dispensing, alarm, inventory, and system events.

Traces to: FR-HSD-01, FR-HSD-02, FR-HSD-03, FR-HSD-04, NFR-TRA-01, NFR-TRA-02

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Retrieve Historical Data (Normal)	Historical ADC event records exist.	An authorized user requests historical records.	The matching historical records are displayed with their associated event information.
Filter Historical Data (Normal)	Historical records exist in the monitoring system.	The user applies filters based on date, time, event type, or ADC.	Only records matching the selected filters are displayed within 5 seconds.
Unauthorized Access (Off-Normal)	A user does not have permission to access historical records.	The user attempts to retrieve historical data.	Access is denied and the historical records are not displayed.

5. Reporting

US-REP-01 — Generate Reports

User Story:
As a hospital administrator, I want to generate reports from historical and inventory data so that I can review ADC activity and medication stock information.

Traces to: FR-REP-01, FR-REP-02, FR-REP-03

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Generate Filtered Report (Normal)	Historical data is available and the user is authorized to generate reports.	The user selects valid filters and requests a report.	A report containing records matching the selected filters is generated.
Generate Inventory Report (Normal)	Inventory information is available for the selected ADC.	An authorized user requests an inventory report.	The report displays medication stock levels and identifies medications requiring refill or reorder.
Report Generation Recorded (Normal)	A report has been successfully generated.	Report generation is completed.	The requesting user and generation timestamp are recorded and a copy of the generated report is stored.

Unauthorized Report Request (Off-Normal)	A user does not have permission to generate reports.	The user attempts to generate a report.	The report is not generated and access is denied.

6. Operator Dashboard

US-OPD-01 — Monitor Hospital ADCs

User Story:
As a hospital operator, I want a centralized overview of ADC status and active alarms so that I can quickly identify cabinets requiring attention.

Traces to: FR-OPD-01, FR-OPD-02, FR-OPD-03, NFR-USE-01, NFR-USE-02, NFR-USE-03

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
View ADC Overview (Normal)	Configured ADCs are being monitored.	The operator opens the dashboard.	The operational status of each configured ADC is displayed.
Active Alarm Displayed (Off-Normal)	An ADC has an active alarm.	The operator views the dashboard.	The alarm displays its severity, type, ADC number, location, and notified personnel.
View ADC or Alarm Details (Normal)	An ADC or alarm is displayed on the dashboard.	The operator selects the ADC or alarm.	Additional information about the selected ADC or alarm is displayed.
ADC Unavailable (Off-Normal)	Communication with an ADC has been lost.	The operator views the dashboard.	The ADC is visibly identified as unavailable using a text label or visual symbol and not by colour alone.

7. Cabinet Health Monitoring

US-CHM-01 — Detect Cabinet Failure

User Story:
As a technician, I want to know when an ADC loses communication or reports a hardware fault so that I can identify and investigate unavailable cabinets.

Traces to: FR-CHM-01, FR-CHM-02, NFR-SAF-02, NFR-REL-01

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Normal Cabinet Operation (Normal)	A configured ADC is connected and sending expected status messages.	The monitoring system checks the cabinet status.	The ADC is displayed as available with its current operational status.
Communication Lost (Off-Normal)	An ADC is being monitored.	The ADC stops sending its expected status messages.	The ADC is displayed as unavailable and its displayed information indicates that it may no longer be current.
Hardware Fault (Off-Normal)	An ADC is connected to the monitoring system.	The ADC reports a hardware fault.	The affected ADC is displayed with the hardware fault and identified as requiring technician attention.
Other ADCs Remain Available (Fault Isolation)	One configured ADC becomes unavailable.	Other configured ADCs remain connected.	Status and event information from the remaining ADCs continues to be received and displayed.

US-CHM-02 — Recover Cabinet Monitoring

User Story:
As a technician, I want monitoring to recover when an ADC becomes available again so that the cabinet can return to normal monitored operation without unnecessary manual intervention.

Traces to: FR-CHM-03, NFR-AVA-02

Acceptance Criteria:

Scenario	GIVEN	WHEN	THEN
Communication Restored (Normal)	An ADC is displayed as unavailable because communication was lost.	Communication with the ADC is successfully re-established.	The ADC's displayed operational status is restored based on the newly received status information.
Temporary Communication Failure (Off-Normal)	Communication with an ADC has been interrupted.	The monitoring system detects the communication failure.	The monitoring system attempts to re-establish

			communication with the affected ADC.
ADC Remains Unavailable (Off-Normal)	Communication with an ADC has been lost.	Communication has not yet been successfully re-established.	The ADC remains displayed as unavailable and its information remains identified as potentially outdated.

High-Level Architecture Diagram

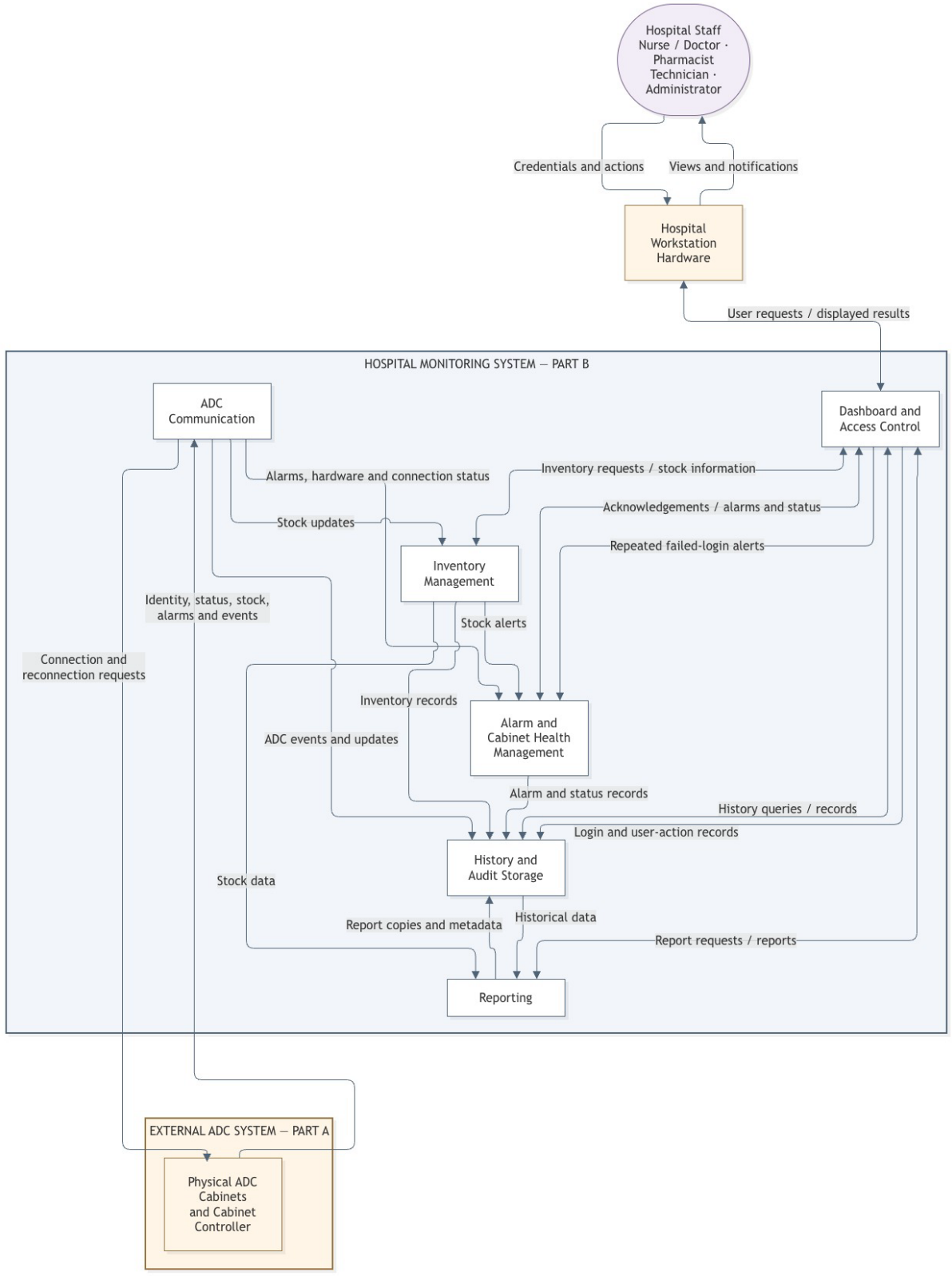


Figure 1: Hospital Monitoring System Architecture

UML Models

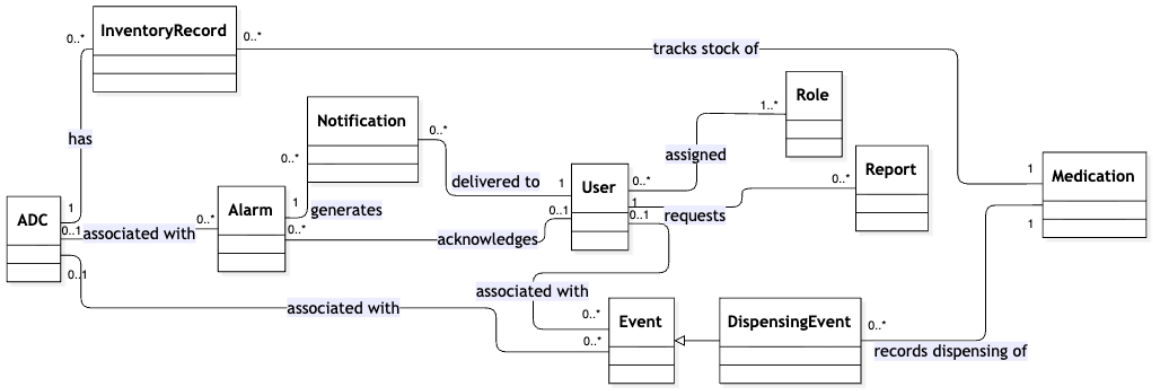


Figure 2: Conceptual Domain Class Diagram

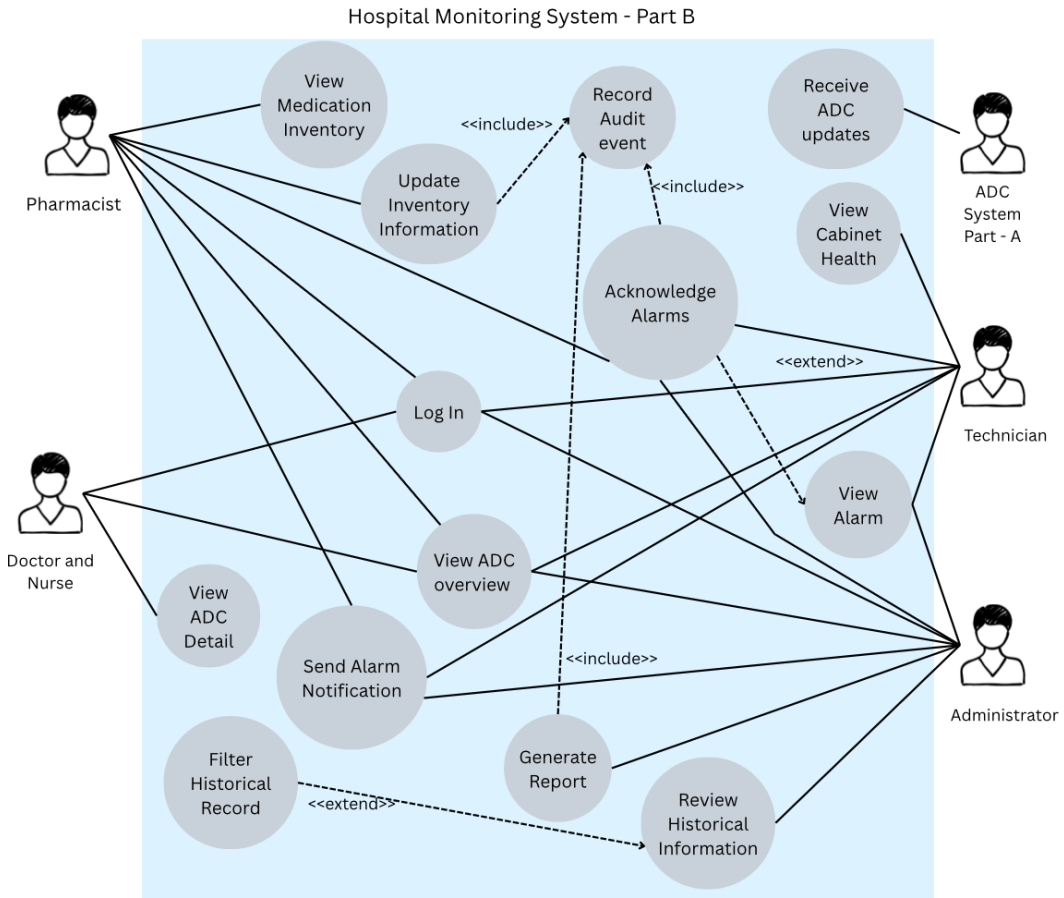


Figure 3: Hospital Monitoring System Use Case Diagram

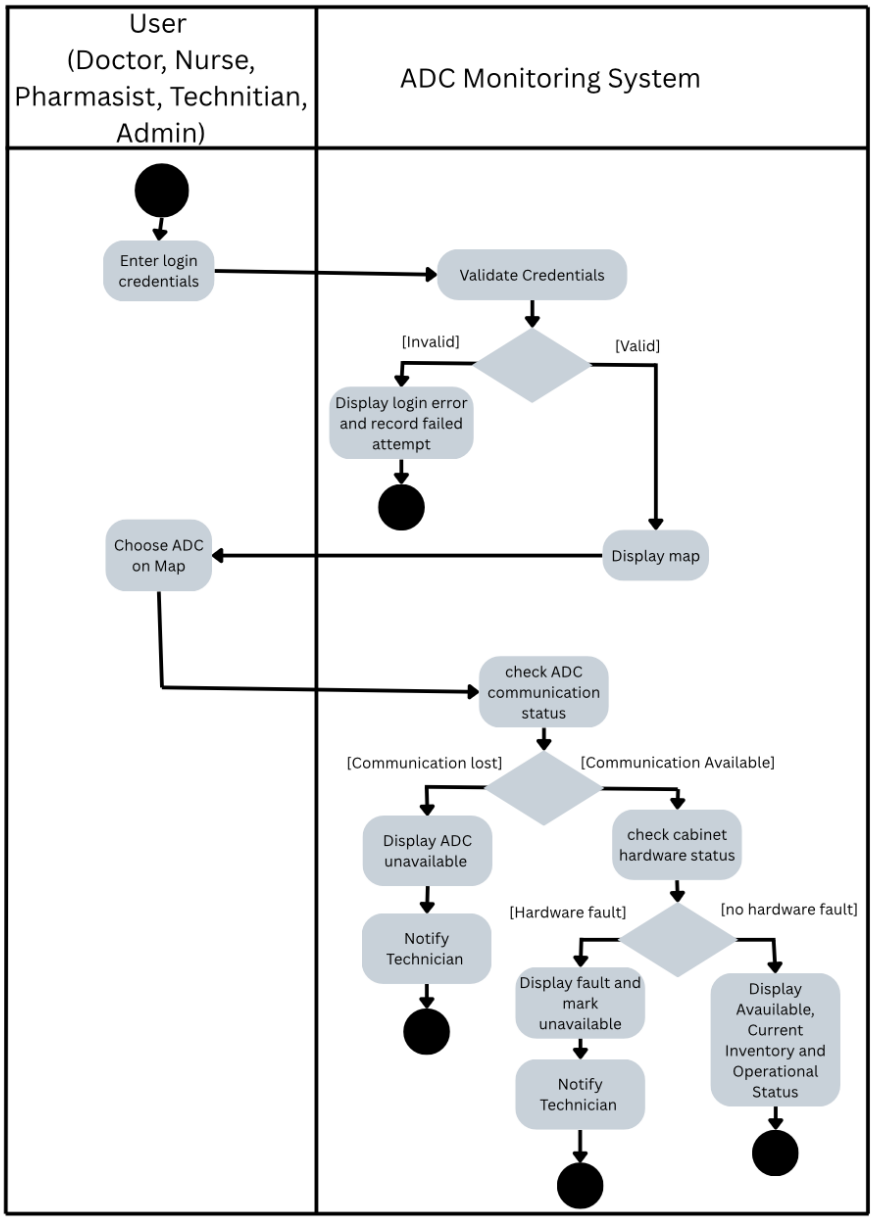


Figure 4: ADC Selection and Status Viewing Activity Diagram

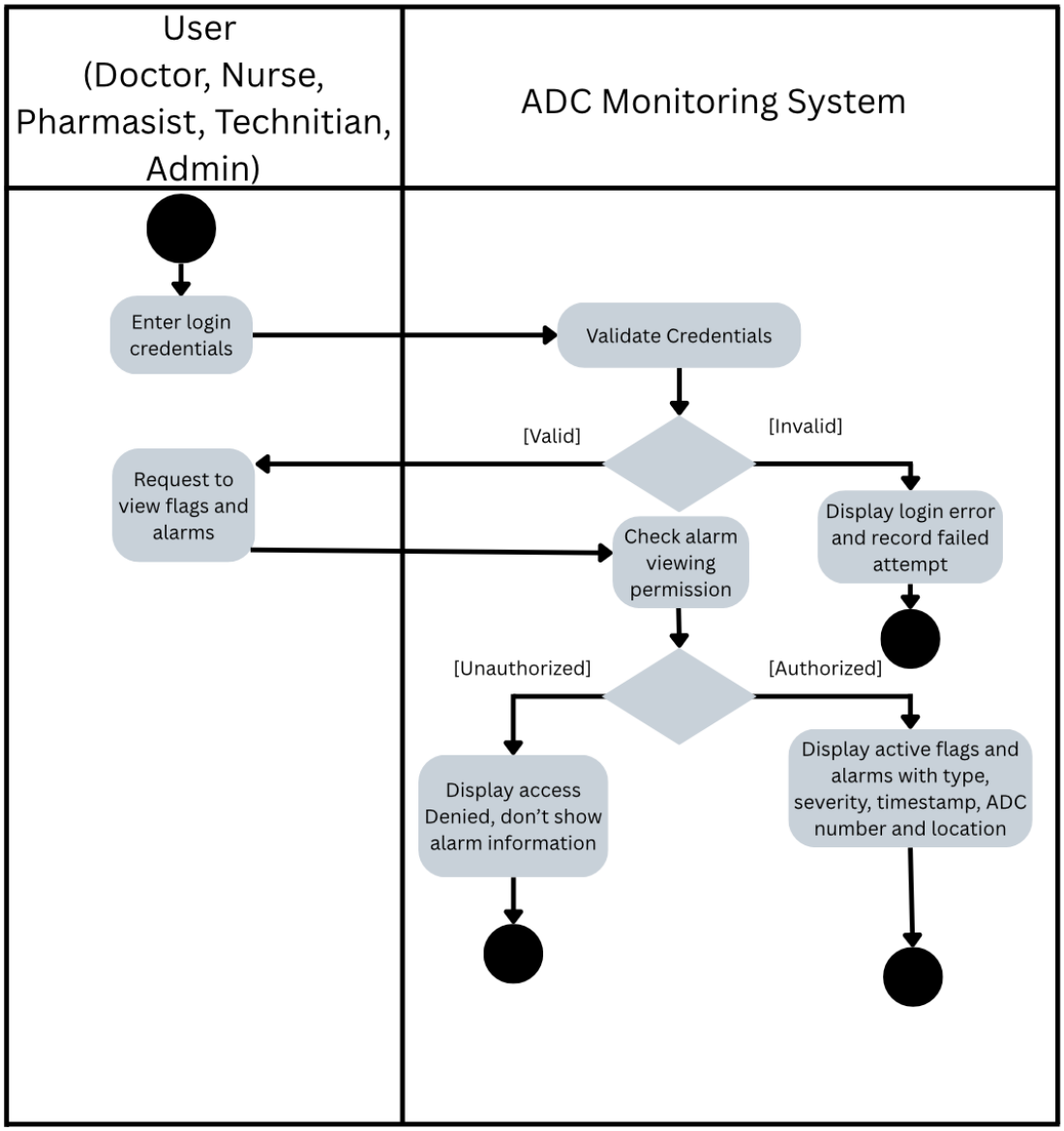


Figure 5: Alarm Viewing Activity Diagram

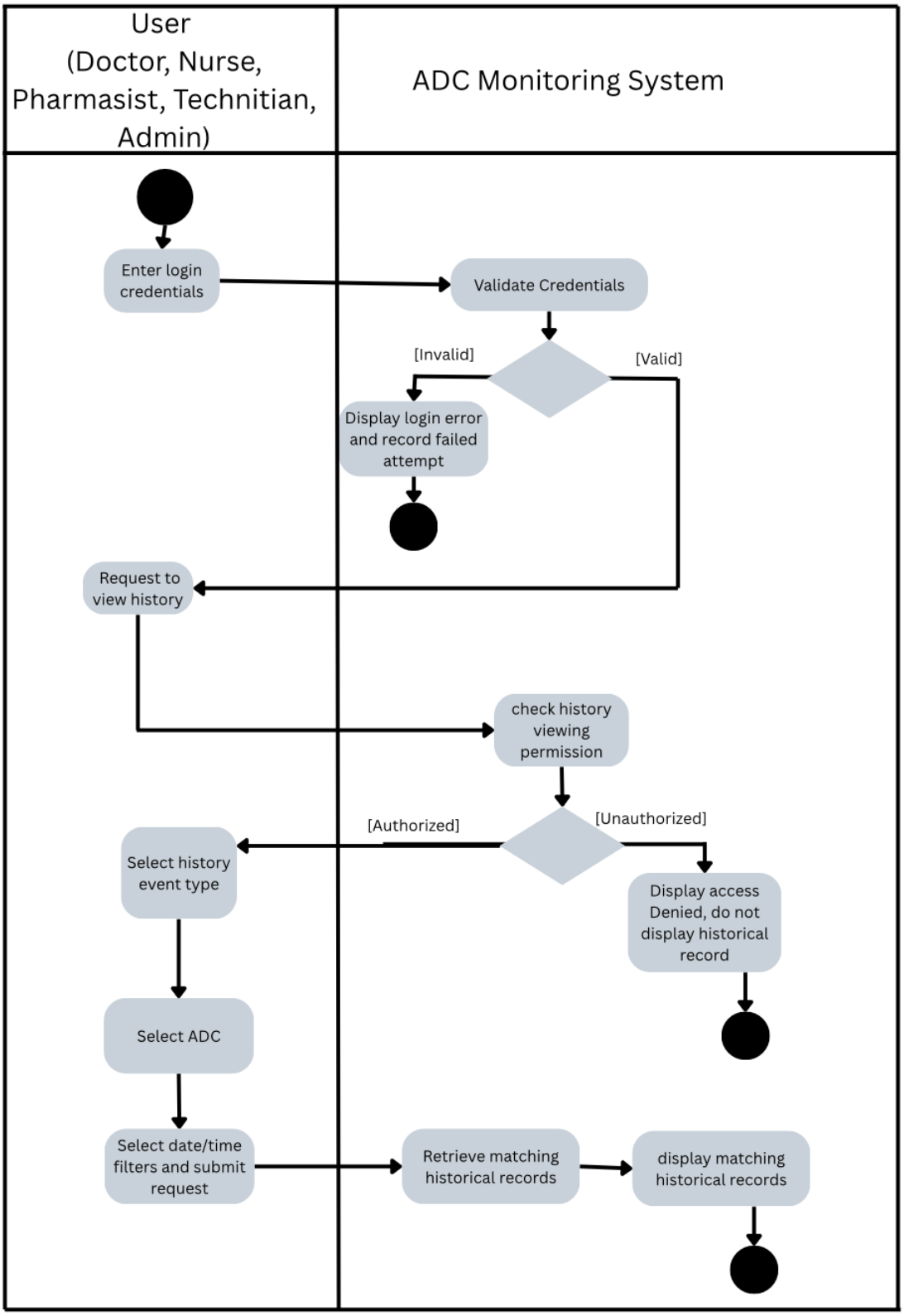


Figure 6: Historical Record Viewing Activity Diagram

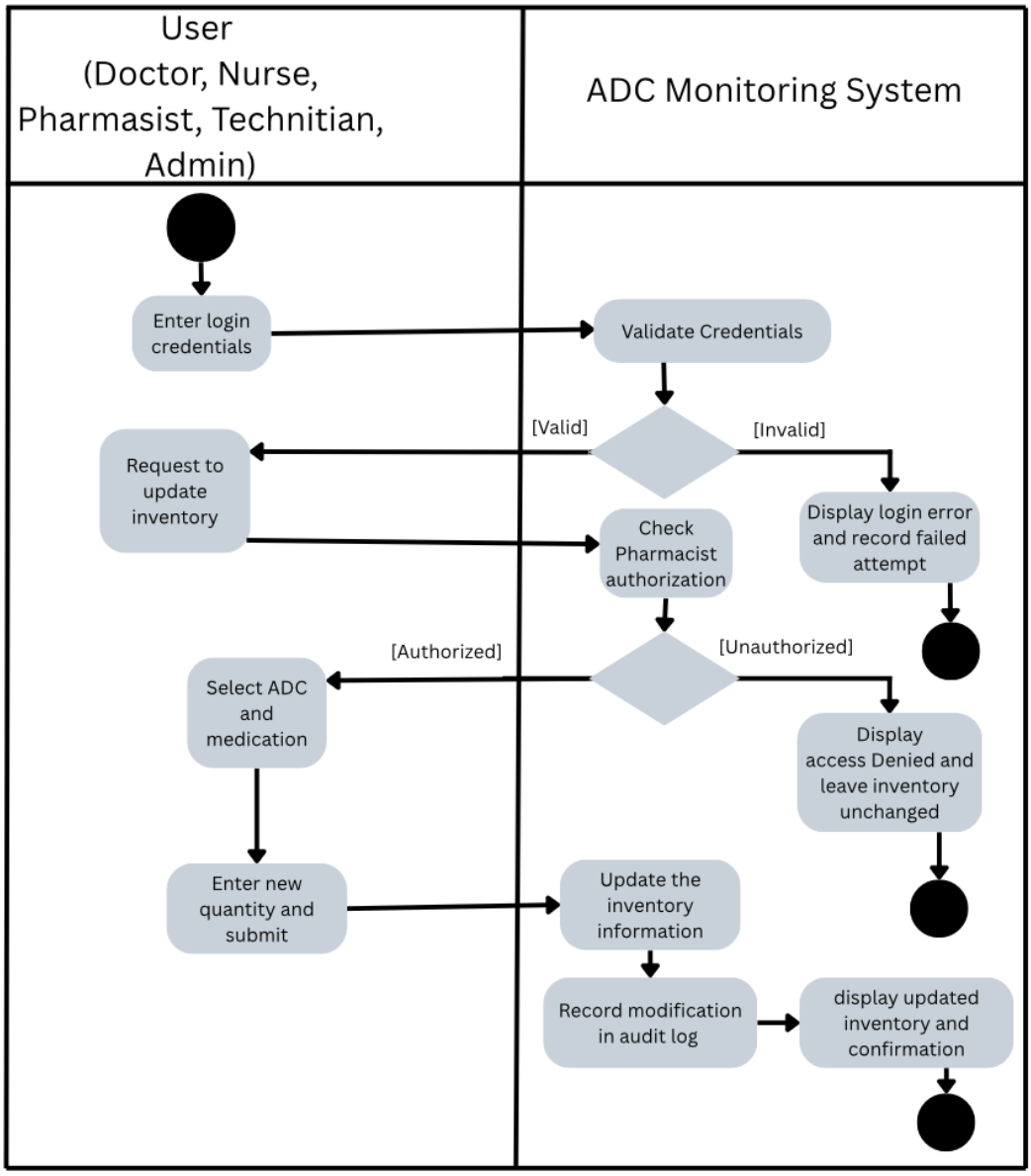


Figure 7: Inventory Update Activity Diagram

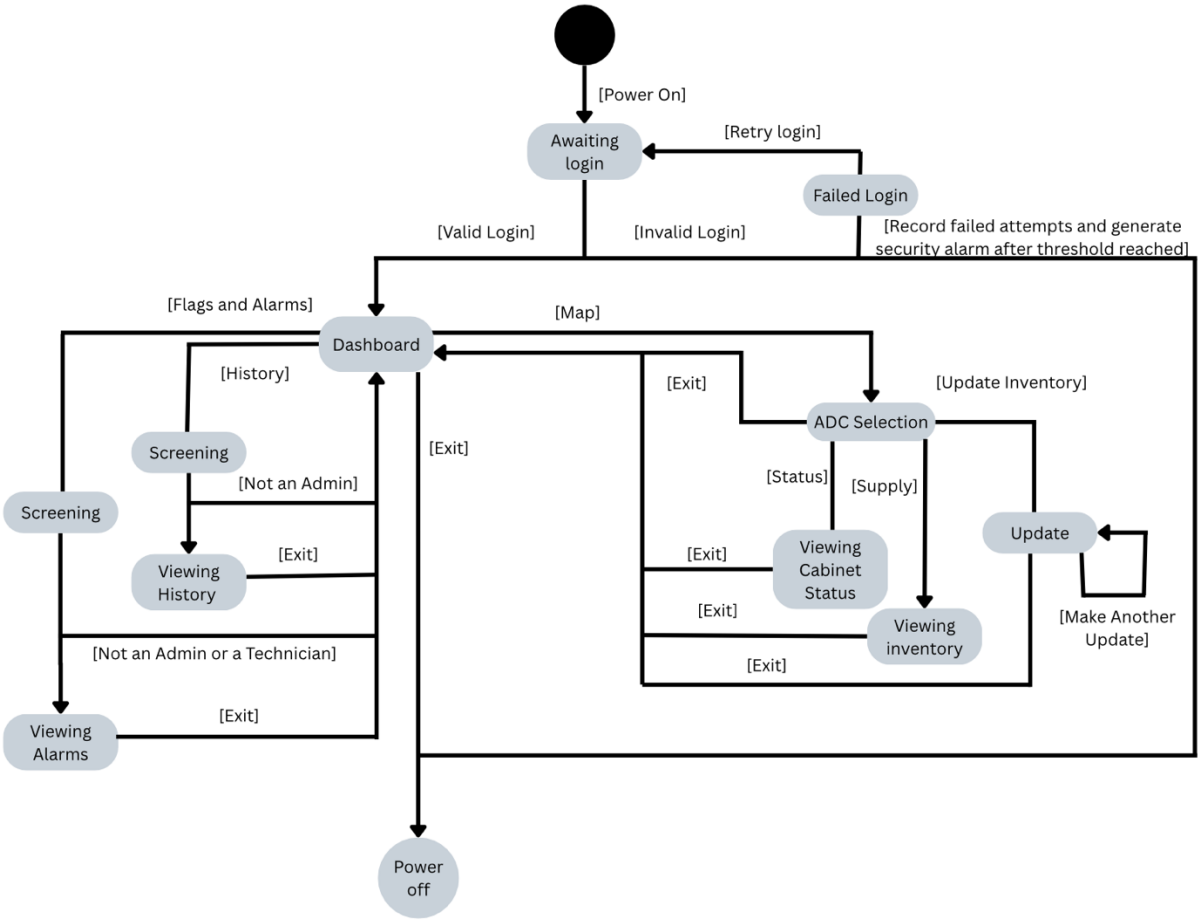


Figure 8: Monitoring Application State Machine Diagram

Traceability Table

Req ID	Requirement (Short)	User Story	Component	Stakeholder / Source
FR-COM-01	Establish and maintain ADC communication	US-COM-01	ADC Communication	Hospital operator; project specification
FR-COM-02	Receive ADC status, inventory, alarm, and event updates	US-COM-01	ADC Communication	Hospital operator; project specification
FR-COM-03	Associate updates with source ADC and location	US-COM-01	ADC Communication	Hospital operator; project specification
FR-INS-01	Monitor medication quantities and stock status	US-INS-01	Inventory Management	Pharmacist; project specification

FR-INS-02	Notify staff when medication requires replenishment	US-INS-01	Inventory Management	Pharmacist; administrative staff
FR-INS-03	Allow authorized inventory viewing and updates	US-INS-02	Inventory Management	Pharmacist; authorized users
FR-ALM-01	Receive, record, and display ADC alarms	US-ALM-01	Alarm and Cabinet Health Management	Hospital operator; project specification
FR-ALM-02	Identify security and operational alarms	US-ALM-01	Alarm and Cabinet Health Management	Administrator; hospital operator
FR-ALM-03	Identify inventory and cabinet-health alarms	US-ALM-01	Alarm and Cabinet Health Management	Pharmacist; technician
FR-ALM-04	Route alarms to appropriate personnel	US-ALM-02	Alarm and Cabinet Health Management	Pharmacist; technician; administrator
FR-HSD-01	Store user login and dispensing event data	US-HSD-01	History and Audit Storage	Administrator; authorized users
FR-HSD-02	Store ADC updates with timestamp and event type	US-HSD-01	History and Audit Storage	Administrator; project specification
FR-HSD-03	Retrieve historical dispensing events	US-HSD-01	History and Audit Storage	Administrator; authorized users
FR-HSD-04	Filter historical data	US-HSD-01	History and Audit Storage	Administrator; authorized users
FR-REP-01	Generate reports from historical data	US-REP-01	Reporting	Administrator
FR-REP-02	Generate medication inventory reports	US-REP-01	Reporting	Administrator; pharmacist
FR-REP-03	Record and retain generated reports	US-REP-01	Reporting	Administrator; audit requirements
FR-OPD-01	Display operational status of ADCs	US-OPD-01	Dashboard and Access Control	Hospital operator
FR-OPD-02	Display active alarms and associated information	US-OPD-01	Dashboard and Access Control	Hospital operator
FR-OPD-03	Allow users to view ADC and alarm details	US-OPD-01	Dashboard and Access Control	Hospital operator

FR-CHM-01	Monitor ADC communication and hardware status	US-CHM-01	Alarm and Cabinet Health Management	Technician
FR-CHM-02	Detect and display unavailable ADCs	US-CHM-01	Alarm and Cabinet Health Management	Technician; hospital operator
FR-CHM-03	Restore cabinet status after recovery	US-CHM-02	Alarm and Cabinet Health Management	Technician

Non-Functional Requirements

Req ID	Requirement (Short)	User Story	Component	Stakeholder / Source
NFR-SAF-01	Clearly distinguish ADC and alarm states	US-ALM-01, US-OPD-01	Dashboard and Access Control	Hospital operator; patient safety
NFR-SAF-02	Indicate unavailable ADC and potentially outdated data	US-COM-01, US-CHM-01	Alarm and Cabinet Health Management	Technician; hospital operator
NFR-SAF-03	Keep active alarms visible until acknowledged or resolved	US-ALM-01	Alarm and Cabinet Health Management	Hospital operator; patient safety
NFR-SEC-01	Require authentication for protected functions	US-INS-02	Dashboard and Access Control	Authorized users; security requirements
NFR-SEC-02	Enforce role-based access control	US-INS-02	Dashboard and Access Control	Nurses; pharmacists; technicians; administrators
NFR-SEC-03	Record repeated failed authentication attempts	US-INS-02	Dashboard and Access Control	Administrator; security requirements
NFR-SEC-04	Lock inactive authenticated sessions	US-INS-02	Dashboard and Access Control	Security requirements
NFR-REL-01	Isolate individual ADC failures	US-CHM-01	ADC Communication	Technician; hospital operator
NFR-REL-02	Preserve records following restart or shutdown	US-CHM-02	History and Audit Storage	Administrator; reliability requirement

NFR-REL-03	Restore stored state following recovery	US-CHM-02	History and Audit Storage; ADC Communication	Hospital operator; reliability requirement
NFR-AVA-01	Remain operational when individual ADCs are unavailable	US-CHM-01	ADC Communication	Hospital operator
NFR-AVA-02	Attempt ADC reconnection after communication failure	US-CHM-02	ADC Communication	Technician
NFR-AVA-03	Be accessible from authorized hospital workstations	US-OPD-01	Dashboard and Access Control	Hospital users; project environment
NFR-PER-01	Display critical alarms within 5 seconds	US-ALM-01	Dashboard and Access Control	Hospital operator
NFR-PER-02	Update ADC status and inventory within 5 seconds	US-COM-01	Dashboard and Access Control	Hospital operator
NFR-PER-03	Send medication notifications within 1 hour	US-INS-01	Inventory Management	Pharmacist; administrative staff
NFR-PER-04	Display requested details and history within 5 seconds	US-HSD-01, US-OPD-01	Dashboard and Access Control	Authorized users
NFR-TRA-01	Maintain audit log of significant user/system actions	US-INS-02, US-HSD-01, US-ALM-01, US-REP-01	History and Audit Storage	Administrator; audit requirements
NFR-TRA-02	Store required information with each audit record	US-HSD-01	History and Audit Storage	Administrator; audit requirements
NFR-TRA-03	Protect audit records from unauthorized modification	US-HSD-01	History and Audit Storage	Administrator; audit/security requirements
NFR-USE-01	Use consistent status indicators and labels	US-OPD-01	Dashboard and Access Control	Hospital operator
NFR-USE-02	Do not communicate critical status using colour alone	US-OPD-01	Dashboard and Access Control	Hospital operator; usability/accessibility

NFR-USE-03	Show complete alarm information without extra navigation	US-OPD-01, US-ALM-01	Dashboard and Access Control	Hospital operator
NFR-USE-04	Display informative error messages	US-OPD-01	Dashboard and Access Control	All system users

Assumptions and Decisions

The following assumptions were made where the project specification did not provide sufficient implementation or operational detail. The resulting decisions were used to define and refine the system requirements. Particular consideration was given to patient safety, privacy, authorized access, accessibility, and the reliability of information presented to hospital personnel.

ID	Assumption	Decision / Impact on Requirements
AD-01 — User Roles	The monitoring system will be used by hospital personnel with different responsibilities, including nurses, pharmacists, technicians, and administrators.	Role-based access control will be used so that users can access only functions appropriate to their assigned role. This influenced the authentication and authorization requirements.
AD-02 — ADC Identification	Each configured ADC can be uniquely identified and associated with a known hospital location.	ADC information and events will be associated with a cabinet number, floor, and wing so users can identify the source and location of an event.
AD-03 — Inventory Thresholds	Medication refill and reorder thresholds will be configured rather than determined by the monitoring system itself.	The monitoring system will monitor received quantities against configured thresholds and notify appropriate personnel when a threshold is reached.
AD-04 — Communication Loss	Information received from an ADC may become outdated when communication with that cabinet is lost.	A cabinet with lost communication will be identified as unavailable, and its displayed information will indicate that it may no longer be current.
AD-05 — Alarm Responsibility	Different types of alarms require responses from different hospital personnel.	Medication-related alarms will be directed to pharmacists, hardware or communication faults to technicians, and security

		or operational alarms to administrators.
AD-06 — Historical Records and Auditability	Hospital personnel may need to investigate previous system activity and determine who performed specific actions.	Significant events and user actions will be recorded with identifying information such as event type, timestamp, user, and affected ADC where applicable.
AD-07 — Patient Safety	Failure to notice a critical alarm or unavailable ADC could contribute to unsafe clinical decisions.	Critical alarms, warnings, unavailable cabinets, and normal states will be visually distinguishable. Active alarms will remain visible until acknowledged or resolved.
AD-08 — Privacy and Controlled Access	Medication, dispensing, user activity, and related hospital records may contain sensitive information and should not be accessible to unauthorized users.	Authentication and role-based access restrictions will be required for protected information and functionality. Audit records will also be protected from modification by normal users.
AD-09 — Accessibility	Users may have difficulty distinguishing information communicated using colour alone.	Critical system states will use text labels or visual symbols in addition to colour. Consistent indicators will be used for normal, warning, critical, and unavailable states.
AD-10 — Failure Isolation	A failure involving one ADC should not make information from otherwise operational ADCs unavailable.	The monitoring system will continue receiving and displaying information from available ADCs when another ADC experiences a communication or hardware failure.
AD-11 — System Recovery	Temporary communication interruptions and application interruptions may occur during normal operation.	The system will attempt to reconnect to disconnected ADCs and preserve stored records across application restarts or unexpected shutdowns.
AD-12 — Performance	Time-sensitive information must reach users quickly enough to support monitoring activities.	Critical alarms and received ADC status information will be displayed within 5 seconds. Medication refill or reorder notifications will be sent within 1 hour of the qualifying inventory update.

Accessibility and Ethical Considerations

Patient Safety:
The monitoring system is intended to support hospital personnel rather than replace clinical judgment. Safety-relevant information, including critical alarms, unavailable ADCs, and potentially outdated information, must therefore be clearly presented to users. The system should avoid presenting stale or unavailable information in a way that could be mistaken for current cabinet status.

Privacy and Confidentiality:
Medication, dispensing, user activity, and other hospital records may contain sensitive information. Access to protected information will therefore require authentication and will be restricted according to the user's authorized role.

Accountability and Auditability:
Actions that may affect medication information, alarm handling, or system records will be traceable to the responsible user where applicable. Audit information will be retained to support investigation and accountability and protected against modification by ordinary users.

Accessibility:
Safety-critical information will not rely exclusively on colour. Text labels and/or visual symbols will accompany status indicators so that important information remains distinguishable for users who may have colour-vision deficiencies.

Human Oversight:
The monitoring system provides information, warnings, and notifications to support hospital personnel. Decisions involving medication management, maintenance, security response, and other clinical or operational actions remain the responsibility of authorized hospital personnel.